

# K8s Attack & Remediation Platform

## Security Assessment Report

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|                               |     |
|-------------------------------|-----|
| Total Attacks Executed        | 10  |
| Critical Severity             | 5   |
| High Severity                 | 4   |
| Completed Successfully        | 10  |
| Failed                        | 0   |
| Detection Alerts Triggered    | 100 |
| Remediation Actions Taken     | 0   |
| MITRE ATT&CK; Tactics Covered | 10  |

# 1. Executive Summary

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This security assessment report documents the execution of 10 real-world Kubernetes attack techniques against a dedicated attack cluster. The assessment was conducted using the K8s Attack & Remediation Platform, which combines automated attack execution with real-time detection monitoring and AI-powered remediation via Claude Sonnet 4. The goal was to evaluate the cluster's security posture by simulating adversarial behaviors mapped to the MITRE ATT&CK; for Containers framework.

## Key Findings

- 5 critical-severity security issues identified, requiring immediate remediation.
- 4 high-severity issues identified, posing significant risk to cluster security.
- 10 MITRE ATT&CK; tactics covered across 10 distinct attack techniques.
- 0 automated remediation actions executed via AI agent with step-by-step reasoning.

**Overall Risk Score: CRITICAL**

## 2. Attack Summary

The following table summarizes all attacks executed during the assessment, including their severity rating, status, and mapped MITRE ATT&CK; technique.

| #  | Attack                                  | Severity | Status    | MITRE Technique |
|----|-----------------------------------------|----------|-----------|-----------------|
| 1  | Privilege Escalation via HostPath Mount | Critical | Completed | —               |
| 2  | RBAC Privilege Escalation               | Critical | Completed | —               |
| 3  | Sidecar Proxy Injection                 | High     | Completed | —               |
| 4  | Container Escape via Privileged Mode    | Critical | Completed | —               |
| 5  | Kubernetes Secrets Exfiltration         | Critical | Completed | —               |
| 6  | ConfigMap Data Collection               | Medium   | Completed | —               |
| 7  | Internal Cluster Network Scan           | High     | Completed | —               |
| 8  | Cluster Resource Hijacking              | High     | Completed | —               |
| 9  | Kubelet API Abuse                       | Critical | Completed | —               |
| 10 | DNS-Based Data Exfiltration             | High     | Completed | —               |

## 3. MITRE ATT&CK; Coverage

The following matrix maps the executed attacks to the MITRE ATT&CK; for Containers framework, demonstrating coverage across multiple tactics and techniques.

| Tactic ID | Tactic Name          | Techniques | Covered Techniques                                       |
|-----------|----------------------|------------|----------------------------------------------------------|
| TA0001    | Initial Access       | 2          | Container and Resource Discovery, Implant Internal Image |
| TA0002    | Execution            | 2          | Container Administration Command, Deploy Container       |
| TA0003    | Persistence          | 2          | Web Shell, Container Escape                              |
| TA0004    | Privilege Escalation | 2          | Escape to Host, Sudo and Sudo Caching                    |
| TA0005    | Defense Evasion      | 2          | Build Image on Host, Disable or Modify Tools             |
| TA0006    | Credential Access    | 2          | Container Secrets, Access K8s API                        |
| TA0007    | Discovery            | 2          | Network Service Scanning, K8s API Discovery              |
| TA0008    | Lateral Movement     | 2          | Kubernetes API, Deploy Container                         |
| TA0009    | Collection           | 2          | Screen Capture, Process Discovery                        |
| TA0040    | Impact               | 2          | Resource Hijacking, Endpoint Denial of Service           |

## 4. Detailed Attack Log

The following section provides detailed information for each attack executed, including the attack methodology, findings, and impact assessment.

## 4.1 Privilege Escalation via HostPath Mount

**CRITICAL**

**Attack ID:** 5e5fef0c-e137-442c-a24f-f2a8448d8cd4

**Severity:** Critical

**Status:** Completed

**MITRE Technique:** —

**Description:** Creates a pod that mounts the host filesystem via hostPath, allowing escape to the underlying node.

## 4.2 RBAC Privilege Escalation

**CRITICAL**

**Attack ID:** c555b5e1-733f-44d3-ab8f-2386ea06ee2c

**Severity:** Critical

**Status:** Completed

**MITRE Technique:** —

**Description:** Exploits cluster-admin binding or overly permissive RBAC roles to gain full cluster control.

## 4.3 Sidecar Proxy Injection

**HIGH**

**Attack ID:** acb7813f-fd7d-4665-b5fb-31e1553ed52a

**Severity:** High

**Status:** Completed

**MITRE Technique:** —

**Description:** Injects a malicious sidecar container into an existing pod to intercept network traffic and exfiltrate data.

## 4.4 Container Escape via Privileged Mode

**CRITICAL**

**Attack ID:** e79878e9-a054-44a7-9481-018c55232704

**Severity:** Critical

**Status:** Completed

**MITRE Technique:** —

**Description:** Creates a privileged container with hostPID and hostNetwork access to escape container isolation and execute commands on the host node.

## 4.5 Kubernetes Secrets Exfiltration

**CRITICAL**

**Attack ID:** 0d2221c2-c24e-4416-945e-e703248a810f

**Severity:** Critical

**Status:** Completed

**MITRE Technique:** —

**Description:** Enumerates and extracts all secrets across namespaces via the Kubernetes API using a compromised service account.

## 4.6 ConfigMap Data Collection

**MEDIUM**

**Attack ID:** 0145df24-7be4-4cba-8cf5-0192595f5224

**Severity:** Medium

**Status:** Completed

**MITRE Technique:** —

**Description:** Extracts all ConfigMaps across namespaces to collect configuration data, credentials, and application settings.

## 4.7 Internal Cluster Network Scan

**HIGH**

**Attack ID:** 376e00f2-9f52-4bf8-a7d4-86f508351d36

**Severity:** High

**Status:** Completed

**MITRE Technique:** —

**Description:** Deploys a pod that scans the internal Kubernetes service CIDR to discover services, open ports, and potential lateral movement targets.

## 4.8 Cluster Resource Hijacking

**HIGH**

**Attack ID:** 0401ec3c-a0f7-41c8-b244-40f6c2c76320

**Severity:** High

**Status:** Completed

**MITRE Technique:** —

**Description:** Deploys resource-intensive pods across available nodes to simulate cryptominer-style resource hijacking and node resource exhaustion.

## 4.9 Kubelet API Abuse

**CRITICAL**

**Attack ID:** b4c9f220-6270-4e90-a979-55d3bda10a75

**Severity:** Critical

**Status:** Completed

**MITRE Technique:** —

**Description:** Discovers kubelet API endpoints (port 10250) on cluster nodes and attempts to execute commands in running pods without going through the API server, bypassing RBAC and audit logging.

## 4.10 DNS-Based Data Exfiltration

**HIGH**

**Attack ID:** 3c10ae5e-a14a-4556-9da2-6f71054965cd

**Severity:** High

**Status:** Completed

**MITRE Technique:** —

**Description:** Simulates data exfiltration via DNS tunneling by encoding stolen data into DNS queries from a compromised pod, bypassing network monitoring.

## 5. Detection & Alerts

The detection monitor continuously observes the cluster for signs of malicious activity. The following alerts were triggered during the assessment.

**Alert Summary:** 0 total alerts — 0 critical, 0 high, 0 medium, 0 low

| Rule    | Severity | Resource | Timestamp          |
|---------|----------|----------|--------------------|
| Unknown | Critical | —        | 1780859251.0221708 |
| Unknown | Critical | —        | 1780859251.0268512 |
| Unknown | High     | —        | 1780859251.026857  |
| Unknown | High     | —        | 1780859251.02686   |
| Unknown | High     | —        | 1780859251.026863  |
| Unknown | Critical | —        | 1780859251.034127  |
| Unknown | High     | —        | 1780859251.034132  |
| Unknown | High     | —        | 1780859251.034195  |
| Unknown | High     | —        | 1780859251.048927  |
| Unknown | High     | —        | 1780859251.052664  |
| Unknown | High     | —        | 1780859251.0526688 |
| Unknown | High     | —        | 1780859251.05392   |
| Unknown | High     | —        | 1780859251.0539289 |
| Unknown | High     | —        | 1780859251.055101  |
| Unknown | High     | —        | 1780859251.055104  |
| Unknown | High     | —        | 1780859251.056257  |
| Unknown | High     | —        | 1780859251.05626   |
| Unknown | High     | —        | 1780859251.0616121 |
| Unknown | High     | —        | 1780859251.0616179 |
| Unknown | High     | —        | 1780859251.06169   |
| Unknown | High     | —        | 1780859251.061693  |
| Unknown | Critical | —        | 1780859251.063812  |
| Unknown | High     | —        | 1780859251.0638149 |
| Unknown | High     | —        | 1780859251.063818  |
| Unknown | Critical | —        | 1780859251.0638669 |
| Unknown | High     | —        | 1780859251.06387   |
| Unknown | High     | —        | 1780859251.063873  |
| Unknown | Critical | —        | 1780859251.0650449 |
| Unknown | High     | —        | 1780859251.0650492 |
| Unknown | High     | —        | 1780859251.0650508 |
| Unknown | High     | —        | 1780859251.065093  |
| Unknown | High     | —        | 1780859251.065096  |
| Unknown | Critical | —        | 1780859281.057645  |
| Unknown | Critical | —        | 1780859281.0578308 |

| Rule    | Severity | Resource | Timestamp          |
|---------|----------|----------|--------------------|
| Unknown | Critical | —        | 1780859281.0653    |
| Unknown | Critical | —        | 1780859281.067911  |
| Unknown | High     | —        | 1780859281.067917  |
| Unknown | High     | —        | 1780859281.067921  |
| Unknown | High     | —        | 1780859281.0679228 |
| Unknown | Critical | —        | 1780859281.07139   |
| Unknown | High     | —        | 1780859281.071396  |
| Unknown | High     | —        | 1780859281.071538  |
| Unknown | High     | —        | 1780859281.085269  |
| Unknown | High     | —        | 1780859281.088928  |
| Unknown | High     | —        | 1780859281.088941  |
| Unknown | High     | —        | 1780859281.090237  |
| Unknown | High     | —        | 1780859281.090241  |
| Unknown | High     | —        | 1780859281.091446  |
| Unknown | High     | —        | 1780859281.091449  |
| Unknown | High     | —        | 1780859281.092581  |
| Unknown | High     | —        | 1780859281.092584  |
| Unknown | High     | —        | 1780859281.094249  |
| Unknown | High     | —        | 1780859281.0942569 |
| Unknown | High     | —        | 1780859281.09555   |
| Unknown | High     | —        | 1780859281.095556  |
| Unknown | Critical | —        | 1780859281.0978    |
| Unknown | High     | —        | 1780859281.097805  |
| Unknown | High     | —        | 1780859281.097808  |
| Unknown | Critical | —        | 1780859281.097871  |
| Unknown | High     | —        | 1780859281.097874  |
| Unknown | High     | —        | 1780859281.097876  |
| Unknown | Critical | —        | 1780859281.098926  |
| Unknown | High     | —        | 1780859281.098929  |
| Unknown | High     | —        | 1780859281.0989308 |
| Unknown | High     | —        | 1780859281.099881  |
| Unknown | High     | —        | 1780859281.099884  |
| Unknown | Critical | —        | 1780859311.086443  |
| Unknown | Critical | —        | 1780859311.086632  |
| Unknown | Critical | —        | 1780859311.0866618 |
| Unknown | Critical | —        | 1780859311.089314  |
| Unknown | High     | —        | 1780859311.089321  |
| Unknown | High     | —        | 1780859311.089324  |
| Unknown | High     | —        | 1780859311.089327  |
| Unknown | Critical | —        | 1780859311.092017  |
| Unknown | High     | —        | 1780859311.092022  |

| Rule    | Severity | Resource | Timestamp          |
|---------|----------|----------|--------------------|
| Unknown | High     | —        | 1780859311.094209  |
| Unknown | High     | —        | 1780859311.1041331 |
| Unknown | High     | —        | 1780859311.112244  |
| Unknown | High     | —        | 1780859311.112252  |
| Unknown | High     | —        | 1780859311.1138828 |
| Unknown | High     | —        | 1780859311.11389   |
| Unknown | High     | —        | 1780859311.115106  |
| Unknown | High     | —        | 1780859311.11511   |
| Unknown | High     | —        | 1780859311.1163788 |
| Unknown | High     | —        | 1780859311.116383  |
| Unknown | High     | —        | 1780859311.1175292 |
| Unknown | High     | —        | 1780859311.117533  |
| Unknown | High     | —        | 1780859311.118638  |
| Unknown | High     | —        | 1780859311.1186419 |
| Unknown | Critical | —        | 1780859311.120981  |
| Unknown | High     | —        | 1780859311.12099   |
| Unknown | High     | —        | 1780859311.120995  |
| Unknown | Critical | —        | 1780859311.1224341 |
| Unknown | High     | —        | 1780859311.1224399 |
| Unknown | High     | —        | 1780859311.122444  |
| Unknown | Critical | —        | 1780859311.124565  |
| Unknown | High     | —        | 1780859311.1245701 |
| Unknown | High     | —        | 1780859311.124573  |
| Unknown | High     | —        | 1780859311.124622  |
| Unknown | High     | —        | 1780859311.124625  |

## 7. Conclusion

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*This assessment successfully demonstrated 10 Kubernetes attack techniques across 10 MITRE ATT&CK; tactics. The identification of 5 critical and 4 high-severity vulnerabilities indicates that the cluster configuration requires immediate hardening.*

### Recommendations

- Enforce Pod Security Standards (restricted profile) to prevent privileged container deployment.
- Implement RBAC least-privilege policies — avoid cluster-admin bindings for service accounts.
- Disable hostPath volume mounts unless absolutely necessary; use CSI drivers with appropriate policies.
- Store secrets in external secrets management (HashiCorp Vault, AWS Secrets Manager) with automatic rotation.
- Enable audit logging and implement real-time threat detection for rapid incident response.
- Conduct regular security assessments to identify and remediate configuration drift.
- Implement network policies to enforce micro-segmentation and restrict east-west traffic.

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*This report was generated automatically by the K8s Attack & Remediation Platform. The findings reflect the state of the cluster at the time of assessment and should be used as part of a broader security review process.*